

Project name:
Number Guessing Game

NAME OF THE STUDENT: Shyantn Kundu

DEPARTMENT: CSE

SECTION: C

CLASS ROLL NO: 190

ENROLLMENT NO: 12022002003028

SUBJECT - IT WORKSHOP LAB

SUBJECT CODE - PCCCS392



INTRODUCTION

"Umm... Let's Guess a Num !!" - A Number Guessing Game

"Umm... Let's Guess a Num!!", this MATLAB project creates a fun, engaging, and intellectual gaming environment.

I. An Intricate Coding Structure:

"Umm... Let's Guess a Num !!" has a complex and precise coding structure at its core. The coding framework is meticulously crafted to ensure smooth handling of all in-game actions, providing impeccable gameplay.

II. Failproof Input Validation:

The project has a failproof input validation process that ensures error-free gaming. The system validates user inputs for configuring rounds, maximum guessing range, and educated guesses. This guarantees a harmonious gaming experience.

III. Compact Design and Academic Elegance:

The project aims for a simple and user-friendly design. Simplicity and clarity are put at the first priority in the interface, so users can focus on the game's intellectual challenge.

IV. User-Based Conventional UI/UX Design:

This project "Umm... Let's Guess a Num !!" is designed to cater to a broad spectrum of users with a user-centric approach. The interface is designed to be conventional, intellectually oriented, and easily accessible, ensuring an all-encompassing user experience. Familiar design elements have been incorporated that resonate with one's intellectual sensibilities to deliver a seamless and engaging user interface.

ALGORITHM/PSEUDOCODE OF IMPLEMENTATION

BEGIN:

```
INITIALIZE r, m, n2, n3, s1, s2, j.
Set up user interface components.
```

Game Initialization:

```
Observe for a change in the 'ofRoundsSpinner' value.
When the 'ofRoundsSpinner' value changes:
    r <- ofRoundSpinner.Value
ENABLE 'MaxGuessingRangeSpinner'
IF 'r' != valid number
    PRINT ERROR
    RETURN
ELSE
    Observe for a change in the 'MaxGuessingRangeSpinner' value.
    When the 'MaxGuessingRangeSpinner' value changes:
        m <- MaxGuessingRangeSpinner.Value
        ENABLE 'StartGameButton'
        IF 'm' != valid number || m==0
            PRINT ERROR
            RETURN
    END
END
```

Starting the Game:

```
When the 'StartGameButton' is pressed:
    DISABLE 'ofRoundsSpinner' & 'MaxGuessingRangeSpinner'.
    m <- MaxGuessingRangeSpinner.Value
    DISABLE 'StartGameButton'.
    ENABLE 'GamePlayPanel'
    SET j<-1
    ENABLE 'RoundEditField', 'EnteraRandomNumberSpinner', and their labels.
```

During Gameplay:

```
PRINT j % Round#
Observe for changes in the 'EnteraRandomNumberSpinner' value.
When the value changes:
    n2 <- EnteraRandomNumberSpinner.Value %USER GUESS INPUT
    ENABLE necessary components based on input validation.
    If n2>m
        PRINT ERROR and DISABLE 'GuessButton'.
        When the 'GuessButton' is pressed:
            DISABLE 'EnteraRandomNumberSpinner' and 'GuessButton'.
            n3 <- random [0 m] %SYSTEM GUESS INPUT
            PRINT n2 AND n3 %DISPLAYING GUESSES OF USER AND SYSTEM
            IF n2>n3
                s1=s1+1
            ELSE IF n2==n3
                s1=s1+1
                s2=s2+1
            ELSE
                s2=s2+1
        END
        PRINT s1 & s2 %DISPLAYING SCORES OF USER AND SYSTEM
        ENABLE necessary components for displaying scores.
        r <- r-1
```

```

        IF r>0
        ENABLE 'ContinuetonextroundButton' & 'EXITViewScoreButton'
    END

```

Continuing to the Next Round:

When the 'ContinuetonextroundButton' is pressed:

```

        IF r>0
        j <- j+1
    END
    DISABLE 'ContinuetonextroundButton' & 'EXITViewScoreButton'.
    Reset guess fields &
    ENABLE 'EnteraRandomNumberSpinner' for the next round.

```

Viewing Final Scores:

When the 'EXITViewScoreButton' is pressed || r==0:

```

    DISABLE all relevant components
    ENABLE the final scores panel.
    PRINT s1 & s2 % USER & SYSTEM FINAL SCORES
    IF s1>s2
        PRINT "USER is Winner"
        PRINT "SYSTEM is Loser"
    ELSE IF s1==s2
        PRINT "USER & SYSTEM in a TIE"
    ELSE
        PRINT "SYSTEM is Winner"
        PRINT "USER is Loser"
    END

```

Starting a New Game:

When the 'NewGameButton' is pressed:

```

    Reset all variables and user interface components.
    ENABLE 'ofRoundsSpinner' to start a new game.

```

END

PCCCS392, IT WORKSHOP LAB
RESULT



Fig 1. GUI of the game after first Start-up

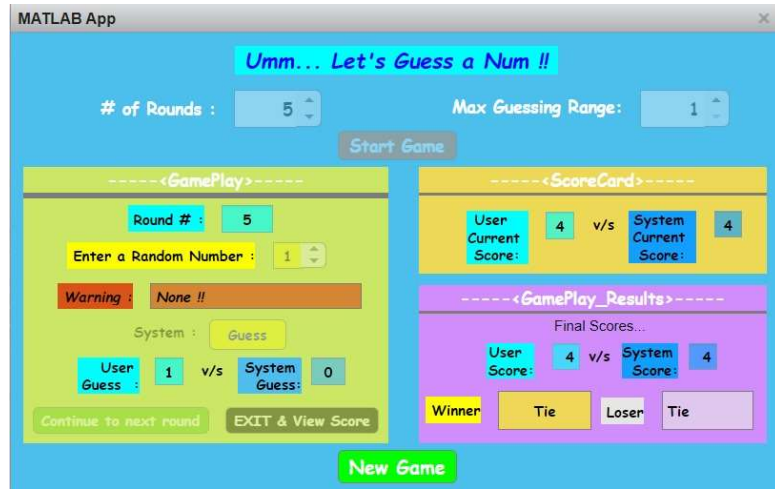


Fig 2. GUI of the game after certain gameplay



Fig 3.1 no. of rounds to be played = 3, maximum guessing range = 2 (from 0), start game button enables



Fig 3.2 start button pressed, round #1, prompts to enter numeric value for guessing



Fig 3.3 after entering random number, system guess button enables

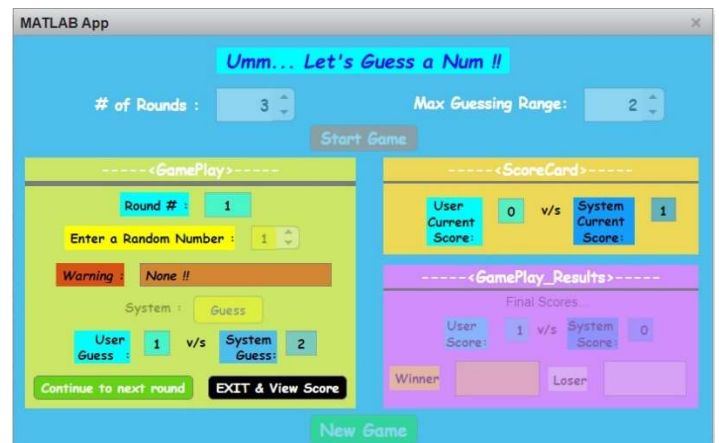


Fig 3.4 after pressing system guess, user and system guesses & scores updates, continue & exit button enables,

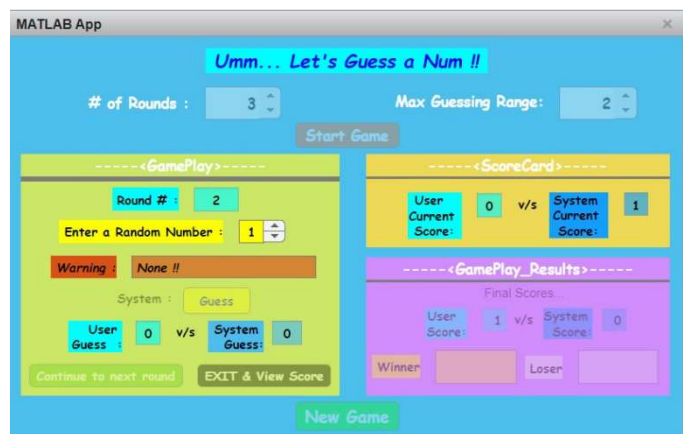


Fig 3.5 continue button pressed, round# updates to 2, User & System guesses resets to 0, user entry for a random number enables, other fields and buttons disabled



Fig 3.6 new user entry made, guess button pressed, new guesses gets updated, scorecard gets updated, continue and exit button enables again



Fig 3.7 exit pressed after round 2, latest score updates into result, winner & loser declared, New game button enables,



Fig 3.8 all fields resets to empty and 0 respectively, # of rounds enables for input entry by user

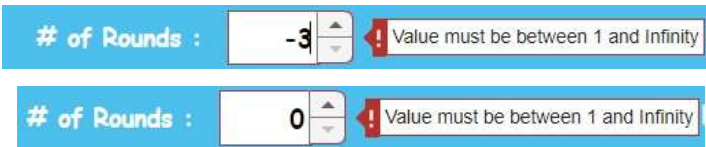


Fig 4.1 Prompts to enter value between 1 and infinity



Fig 4.2 Prompts to enter numeric value



Fig 4.3 Correct entry of the data field “# of Rounds”

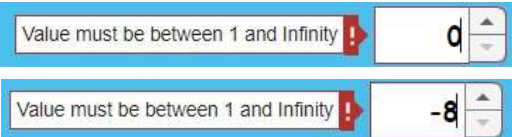


Fig 4.4 Prompts to enter value between 1 and infinity



Fig 4.5 Prompts to enter numeric value



Fig 4.6 Correct entry of the data field “Max Guessing Range”

CONCLUSION

In conclusion, "Umm... Let's Guess a Num !!" is a well-crafted and intellectually inspired number guessing game. The project's coding structure is intricate, which ensures accuracy and robustness in handling user inputs and game mechanics. With failproof input validation, a compact and elegant design, and a user-centric approach, this project aims to deliver an engaging and error-free gaming experience for users of all levels.

I extend my heartfelt gratitude to Prof. Bavrabi Ghosh for her choosing such a unique topic to build an application on, provide support, guidance throughout the project's development. I am also grateful to the entire IT Workshop and MATLAB Lab for their invaluable contributions and assistance.